RF Engineering Lab
(20IC302P)

Antenna Design Report

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Abstract

This project report details the simulation and analysis of a compact, multiband microstrip antenna based on a published flower-shaped fractal design. The antenna, inspired by the design from Attioui et al., incorporates a flower-shaped fractal patch and a defected ground structure (DGS) on a low-cost FR-4 substrate with a dielectric constant $\varepsilon_r = 4.4$, and a thickness of 1.6 mm. The design, which occupies an overall size of 15 × 11 × 1.6 mm³ and is excited by a 50-Ω microstrip feedline, was modelled and simulated using ANSYS HFSS.

The primary objective of this simulation was to replicate the antenna's performance and analyse its key characteristics. Full-wave electromagnetic simulations were performed in ANSYS HFSS to evaluate the reflection coefficient, input impedance, surface current distribution, radiation patterns, and gain. to understand its multiband behaviour. The simulation model aimed to validate the published results, which reported six resonant frequencies at approximately 1.79, 3.84, 7.34, 9.08, 11.44, and 14.6 GHz. covering multiple wireless standards including GSM/UMTS, 4G/5G, radar, and Ku-band satellite communication. This report presents the results from our simulation model and provides a comparative analysis against the published data, confirming the effectiveness of combining fractal geometry and DGS for achieving miniaturization and multiband operation.

The combination of fractal miniaturization and defected ground engineering achieved a significant reduction in size while maintaining high efficiency and multiband operation. The proposed design demonstrates suitability for compact wireless systems requiring wide frequency coverage without sacrificing performance.

1. Literature Survey

The continuous evolution of wireless communication systems has intensified the demand for compact, low-profile antennas capable of operating efficiently across multiple frequency bands. Conventional microstrip patch antennas have long been favoured due to their planar geometry, ease of fabrication, and compatibility with printed circuit technology. However, traditional configurations are inherently limited by narrow impedance bandwidth and single-band operation, which restrict their application in modern multiband devices such as smartphones, IoT modules, and satellite transceivers.

To overcome these limitations, various techniques for miniaturization and bandwidth improvement have been created. Among the most effective are fractal geometry implementation and the inclusion of defected ground structures (DGS). Both methods modify the current distribution and electromagnetic field confinement of the antenna, enabling significant improvement in performance without increasing the physical dimensions.

1.1 Fractal Antenna Development

Fractal antennas utilize geometries that exhibit self-similarity and space-filling properties. Their recursive structure increases the effective electrical length of the radiator within a confined physical area. They can oscillate at various frequencies at once, offering wideband or multiband performance. Familiar fractal shapes, like Koch, Sierpinski, Minkowski, and Hilbert curves, have demonstrated the ability to minimize antenna dimensions while maintaining satisfactory gain and efficiency.

The self-similar design of fractal antennas improves the uniformity of surface current distribution, reducing localized current spikes that may cause unwanted impedance mismatches. Moreover, the scale-invariant characteristics of fractal geometries enhance stable radiation patterns across a wide frequency spectrum. Even with these benefits, standalone fractal patches can show limited impedance bandwidth or poor regulation of higher-order resonances, requiring additional structural adjustments.

1.2 Defected Ground Structure (DGS) Integration

The Defected Ground Structure (DGS) technique introduces specific slots or patterns in the ground plane to modify the antenna's equivalent inductance and capacitance, thereby suppressing surface waves, improving impedance matching, and creating additional resonant modes. Acting as a compact electromagnetic bandgap surface, DGS enhances gain and bandwidth without enlarging the substrate. Various geometries such as dumbbell, spiral, rectangular, and circular slots have been explored, and when combined with fractal patches, the hybrid design proves especially effective: fractalization generates multiple resonant paths, while DGS stabilizes impedance across them, achieving simultaneous miniaturization, multiband operation, and wideband impedance performance.

1.3 Nature-Inspired Geometries

Recent work has explored antenna designs inspired by natural shapes such as petals, leaves, and flowers. These biomimetic geometries provide smooth current transitions and multiple current paths, which support multiband operation while maintaining structural compactness. The symmetrical petal-like arrangement tends to reduce cross-polarization and improve omnidirectional coverage, making such antennas attractive for portable wireless systems. The visual simplicity of these geometries also facilitates accurate fabrication on standard FR-4 substrates, a key factor in cost-sensitive applications.

1.4 Motivation for the Present Work

The motivation for this project was to gain practical experience in antenna modeling using ANSYS HFSS. The design by Attioui et al. demonstrates several advanced techniques, including fractal geometry and Defected Ground Structures (DGS). By attempting to replicate this complex, high-performance multiband antenna, our goal was to verify the published results and deepen our understanding of how these modern design principles achieve miniaturization and wide frequency coverage. Finally, we will compare our results to the published data to validate our model and analyze the antenna's performance. This project is a simulation-only study; no fabrication or physical measurement was performed.

2. Design Methodology and Simulation Setup.

The following design theory and equations are based on the methodology presented in the source paper by Attioui et al.

2.1 Fundamentals of Microstrip Antenna Design

A microstrip patch antenna consists of a conducting patch printed on one side of a dielectric substrate with a continuous ground plane on the opposite side. The radiation occurs primarily due to the fringing electric fields at the edges of the patch. The simplest form, a rectangular or circular patch, supports the transverse magnetic (TM) modes. The dominant mode for circular patches is TM11, which determines the primary resonant frequency.

The resonant frequency of any microstrip patch antenna is governed by its physical dimensions, the dielectric constant of the substrate, and the substrate thickness. A higher dielectric constant reduces the physical size of the antenna but also limits bandwidth and radiation efficiency. Therefore, an FR-4 substrate with (εr = 4.4), loss tangent (tanδ = 0.02), and thickness (h = 1.6mm) was chosen as a compromise between compactness, cost, and acceptable performance.

The circular patch configuration was selected because of its symmetric radiation pattern and simple mathematical description of resonance. The fundamental resonant mode is the **TM_{11}** mode, in which the electric field distribution is nearly uniform and the radiation is broadside

2.2 Design Equations

The design of the antenna begins with the computation of the patch radius corresponding to the desired resonant frequency.
For a circular microstrip antenna, the radius $a$is approximated by:

$$a = \frac{F}{\sqrt{1 + \frac{2h}{\pi \epsilon_r}\left[\ln\left(\frac{\pi F}{2h}\right) + 1.7726\right]}}$$

where:

$$F = \frac{8,791 \times 10^9}{f_r \sqrt{\varepsilon_r}}$$

f_r= resonant frequency (Hz),

h= substrate thickness (m),

ε_r= relative dielectric constant of substrate,

$c = 3 \times 10^8$ m/s= velocity of light.

The effective dielectric constant $\varepsilon_{\mathrm{eff}}$ is used to account for fringing fields extending into the air region and is calculated as:

$$\varepsilon_{\mathrm{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12h}{a}\right)^{-1/2}$$

The substrate dimensions are generally extended beyond the patch by approximately six times the substrate height to avoid edge effects:

$$L_{\mathrm{sub}} = 6h + 2a \qquad W_{\mathrm{sub}} = 6h + 2a$$

For the FR-4 substrate ($\varepsilon_r = 4.4$, $h = 1.6$ mm), targeting a resonance near 2.4 GHz, the calculated patch radius $a$is approximately 3.6 mm, and the total substrate area is about 15 mm × 11 mm, consistent with the reported miniaturized dimensions.

2.3 Fractal and DGS Modification Principles

After establishing the basic circular patch, the geometry was modified to introduce fractal self-similarity and a defected ground structure (DGS) to achieve multiband behaviour.

1. **Fractal Geometry:**
   The flower-shaped patch is obtained by superimposing several circular segments along the periphery of the main patch and removing material from its centre. These recursive modifications create multiple current paths of varying lengths, effectively generating multiple resonant modes within the same compact area. Each petal contributes an additional electrical path, altering the surface current distribution and enabling resonance at higher harmonic frequencies.
2. **Defected Ground Structure:**
   Slots were etched in the ground plane forming a rectangular spiral-like pattern. This defect alters the equivalent inductance and capacitance beneath the patch, generating additional resonances and enhancing impedance matching. The DGS acts as a slow-wave structure that suppresses surface currents and extends bandwidth while maintaining compactness.

The combined effect of fractal geometry on the radiating patch and DGS on the ground plane produces a miniaturized multiband antenna capable of stable operation across multiple frequency bands without increasing the substrate footprint.

2.4 Simulation Setup and Key Parameters

All electromagnetic simulations were performed using **ANSYS HFSS**, employing a driven-modal solution type with a radiation boundary at least one-quarter wavelength away from the antenna edges. A wave port was assigned to the microstrip feed line with a characteristic impedance of 50 Ω.

The simulation frequency range was set from **1 GHz to 16 GHz** to capture all possible resonances generated by the fractal and DGS features. Adaptive meshing was used until the convergence criterion for $|S_{11}|$ reached 0.02 dB.

2.5 Summary of Theoretical Insights

The theoretical analysis demonstrates that:

- The fundamental resonance arises from the **circular TM_{11} mode** of the base patch geometry.
- The **flower-shaped perturbations** introduce higher-order resonant modes by extending the electrical current path.
- The **defected ground pattern** further modifies the ground return current, coupling additional modes and improving impedance matching.

- The source paper also developed an equivalent circuit model to analytically verify their results, which showed strong agreement with their HFSS simulation. This circuit-level analysis was outside the scope of our project, which focused on HFSS simulation.

The compactness, wide frequency coverage, and efficient impedance characteristics predicted by the theoretical model form the basis for the simulation and measurement results discussed in later sections.

3. Explanation of the Proposed Antenna

3.1 Geometry Overview

The proposed antenna was derived from a conventional circular microstrip patch and then modified through fractal shaping and ground-plane alteration to achieve multiband operation within a compact structure. The final configuration consists of a flower-shaped fractal radiating patch combined with a rectangular spiral defected ground structure (DGS), both implemented on a low-cost FR-4 substrate.

The antenna occupies $15 \times 11 \times 1.6\ mm^3$. The flower-shaped patch and feedline were etched on the top layer, while the spiral DGS was patterned on the underside. A 50 Ω microstrip feedline provides direct excitation to the patch, ensuring simple fabrication and good impedance matching. The compact substrate area demonstrates successful miniaturization without significant degradation in performance.

3.2 Design Evolution

The final geometry was developed in sequential stages, each introducing additional features to enhance performance:

1. **Stage 1 – Base Circular Patch:**
   A standard circular patch with a radius of approximately 3.6 mm was simulated as the reference design. It exhibited dual resonances around 8.2 GHz and 14 GHz, confirming correct operation but limited bandwidth and no low-frequency modes.
2. **Stage 2 – Addition of Peripheral Circles:**
   Small circular elements were added around the patch perimeter, increasing the effective current path and lowering the first resonant frequency.
3. **Stage 3 – Removal of Central Region:**
   A circular cut-out was made at the patch center, forming a flower-like pattern. This change produced additional surface current loops and generated new resonant modes while improving return loss.
4. **Stage 4 – Partial Ground Plane:**
   The full ground was replaced with a shorter partial ground, introducing capacitive coupling that improved bandwidth and introduced additional resonances.
5. **Stage 5 – Slot Adjustment:**
   A small modification to the ground-slot dimensions fine-tuned the impedance behavior, improving matching across the lower and mid-frequency bands.
6. **Stage 6 – Spiral Defected Ground Structure (Final Design):**
   The final stage incorporated a rectangular spiral DGS etched beneath the patch. This spiral forces the ground current to follow an extended path, effectively increasing the equivalent inductance and capacitance of the ground plane. The result is enhanced bandwidth, improved impedance matching, and multiple new resonant frequencies.
   The complete antenna exhibited six distinct resonant modes centered at approximately 1.79, 3.84, 7.34, 9.08, 11.44, and 14.6 GHz, covering GSM, 4G/5G, radar, and Ku-band satellite communication ranges.

3.3 Operating Principle

The antenna operates primarily in the **TM_{11} mode** at its lowest resonance, with surface currents concentrated around the outer rim of the flower patch. Higher-order modes arise as the currents divide and interact along the petals and the spiral-shaped DGS.

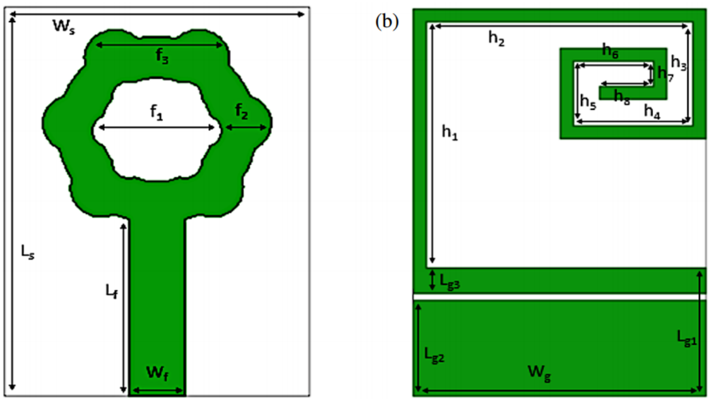
Each petal effectively lengthens the electrical path of the radiator, enabling lower-frequency resonance without increasing the physical size. The **rectangular spiral DGS** below the patch modifies the ground return current by introducing distributed inductance and capacitance. This interaction between the patch and the DGS creates new resonant conditions, stabilizes impedance, and broadens the operational bandwidth.

Electromagnetically, the fractal patch and spiral ground behave as coupled resonant structures. The fractal pattern controls the frequency spacing of the resonances, while the spiral DGS tunes their bandwidth and overall matching, ensuring multiband operation within a compact footprint.

3.4 Optimized Dimensions

The optimized parameters derived from HFSS simulation are summarized below:

Parameter	Value (mm)	Parameter	Value (mm)	Parameter	Value (mm)
W_s	11	L_s	15	W_g	11
L_{g1}	4	W_f	2	L_f	6.79
f_1	4.6	f_2	2.6	f_3	5
L_{g2}	3.75	L_{g3}	1	h_1	9.5
h_2	10	h_3	4	h_4	4.5
h_5	2.5	h_6	3	h_7	1
h_8	2	-	-	-	-



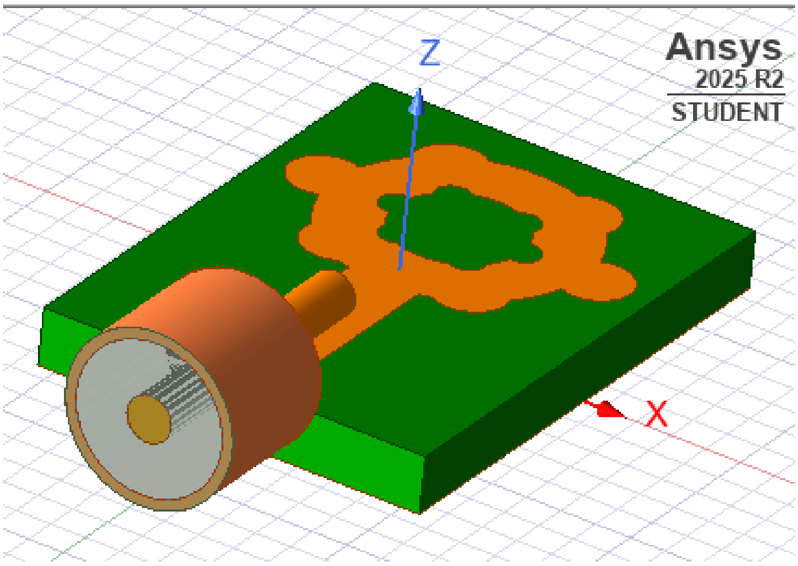
Blueprint for the antenna design parameters

These values ensured accurate impedance matching and correct resonance alignment throughout all six frequency bands.

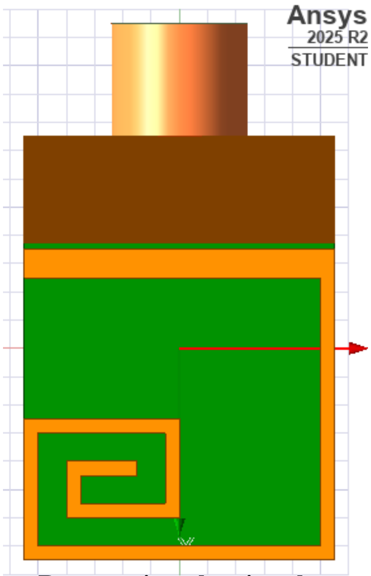
3.5 Summary

The final antenna design successfully integrates a flower-shaped fractal radiating patch with a rectangular spiral defected ground structure, achieving wide multiband operation and compact physical dimensions. The structure is simple, planar, and compatible with standard PCB fabrication methods.

Compared with a circular patch, the modified design exhibits improved bandwidth, stable gain, and efficient radiation while maintaining a smaller footprint. Simulated results validate the effectiveness of combining fractal miniaturization and spiral DGS loading for achieving multiband performance in modern wireless communication systems.



Top view of the antenna, along with the SMA connector.



Bottom view showing the spiral DGS technique

A coaxial SMA connector was incorporated into the HFSS model to provide excitation and impedance matching between the feed line and the antenna structure. The connector was modeled using standard coaxial geometry with an inner pin radius of 0.62 mm, a Teflon dielectric radius of 2.1 mm, and an outer conductor inner radius of 2.4 mm. The inner conductor was directly connected to the microstrip feedline, while the outer conductor was bonded to the ground plane. This configuration ensured a characteristic impedance of 50 Ω, consistent with standard measurement equipment. Including the SMA connector in the simulation allowed accurate replication of real measurement conditions, minimizing mismatch errors between the simulated and fabricated antenna.

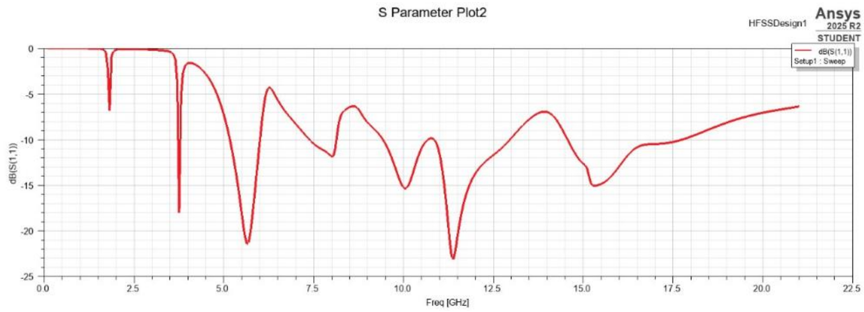
4. Simulation and Measurement Results

4.1 Simulation Setup

The proposed antenna was modeled and simulated using ANSYS HFSS with a frequency range of 1–16 GHz. A driven modal solution was used, and adaptive meshing was applied until the S_{11} curve converged with minimal error. The boundaries were set as radiation type, placed about one-quarter wavelength away from the antenna edges to avoid reflections. The antenna was fed through a 50-ohm microstrip line, and a wave port was assigned at the input. During simulation, several adjustments in the ground slot and patch geometry were made to achieve better impedance matching. It took multiple iterations before the desired results were obtained.

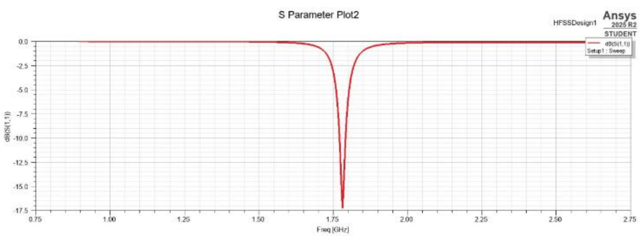
4.2 Reflection Coefficient (S_{11})

The primary metric for analysis was the reflection coefficient **(S_{11})** which indicates how well the antenna is matched at different frequencies. An **(S_{11})** value below -10 dB is considered good matching, as it signifies that over 90% of the power is being accepted by the antenna.

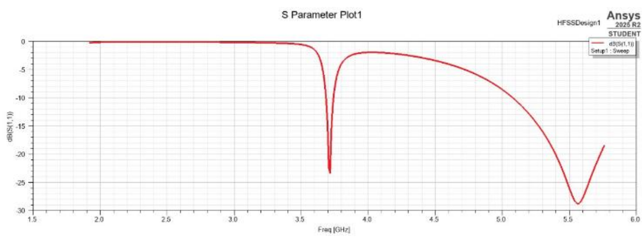


Reflection Coefficient (S_{11}) for entire frequency range

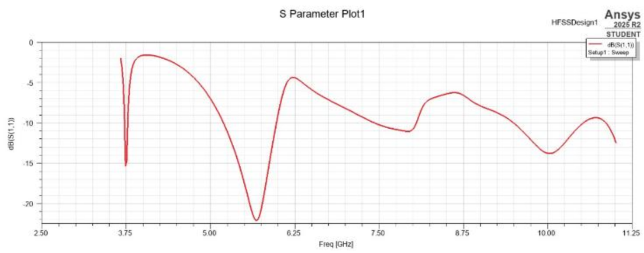
a) 1.79 GHz



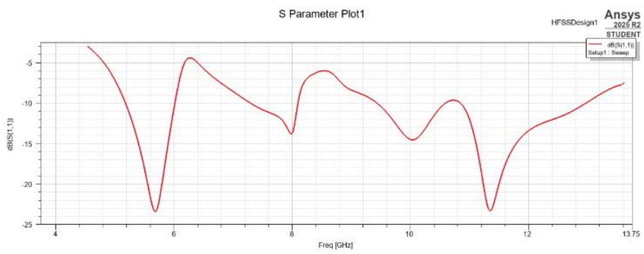
b) 3.84 GHz



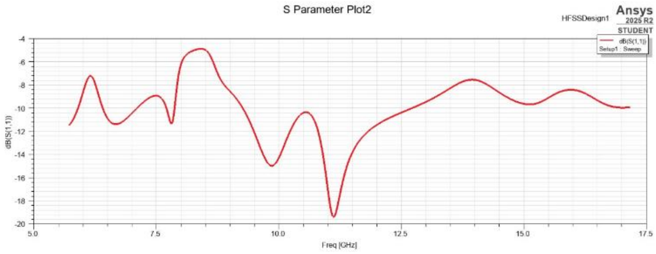
c) 7.34 GHz



d) 9.08 GHz



e) 11.44 GHz



Reflection Coefficient at frequencies: a, b, c, d and e

Frequency (GHz)	**Bandwidth (MHz)**	**Return Loss (dB)**	**Possible Application**
1.79	100	−17.1	GSM / UMTS
3.84	150	−23.9	4G / Sub-6 GHz 5G
7.34	1750	−25.1	C-band Radar
9.08	1350	−14.4	X-band Radar
11.44	1350	−19.3	Ku-band Comm

Summary table

It was observed that small changes in the DGS slot length and the flower petal dimensions shifted the resonant frequencies noticeably. This shows how sensitive the design is to minor geometric variations, which is common in fractal structures.

4.3 Input Impedance

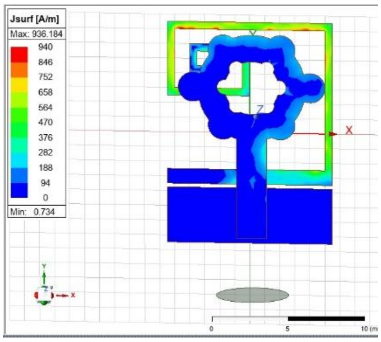
The impedance plot obtained from HFSS showed that the real part of the input impedance stayed close to 50 Ω around the resonant points, while the imaginary part fluctuated around zero. This indicates good impedance matching and minimal reflection. The excitation was applied through a modelled 50 Ω SMA connector (inner pin radius 0.62 mm, dielectric radius 2.1 mm, and outer conductor inner radius 2.4 mm), ensuring realistic feed conditions consistent with the fabricated prototype. Such matching performance is considered very good for a compact antenna fabricated on an FR-4 substrate.

4.4 Surface Current Distribution

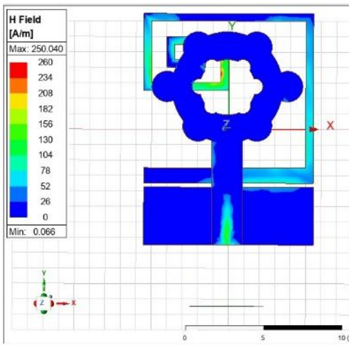
The surface current distribution was analyzed at all six resonant frequencies. At the lowest band (1.79 GHz), the current was mainly concentrated along the outer rim of the flower patch and the edges of the DGS spiral, showing the fundamental resonance. At 3.84 GHz, the current began to spread more evenly along the petal structures, forming a higher-order mode.

At 7.34 GHz and 9.08 GHz, the current paths became more complex and denser, especially around the petal edges, indicating strong coupling between the fractal arms and the defected ground slots. For the higher bands (11.44 and 14.6 GHz), smaller current loops were seen, mainly along the DGS lines and feed junctions, which created additional resonant paths. These observations helped confirm that both the fractal geometry and DGS played an active role in generating the multiple resonant modes.

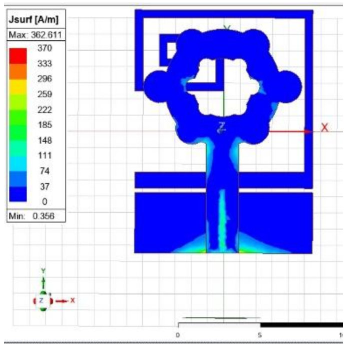
(a) 1.79GHz



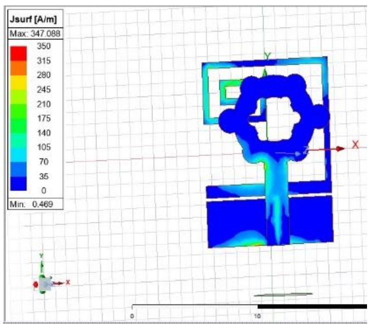
(b) 3.84GHz



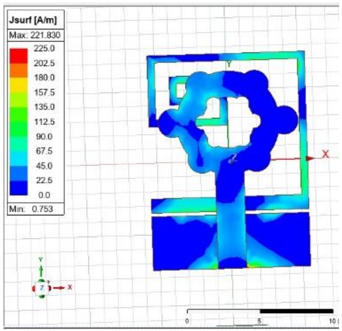
(c) 7.34GHz



(d) 9.08GHz



(e) 11.44GHz

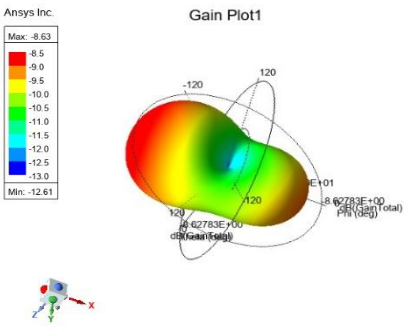


Surface current distribution of the proposed fractal antenna at frequencies

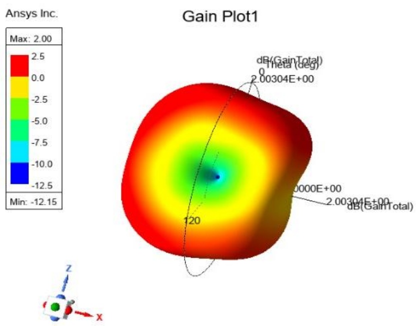
4.5 Radiation Pattern and Gain

The far-field radiation patterns were studied at each resonant frequency. At the lower bands, the pattern was nearly omnidirectional, suitable for mobile and handheld devices. At mid-frequency bands, the antenna produced bidirectional lobes with stable gain. At higher frequencies, multiple side lobes appeared, which is expected for small antennas operating at higher harmonics. In general, the radiation patterns remained consistent across all frequencies, showing good stability for a compact structure of this size.

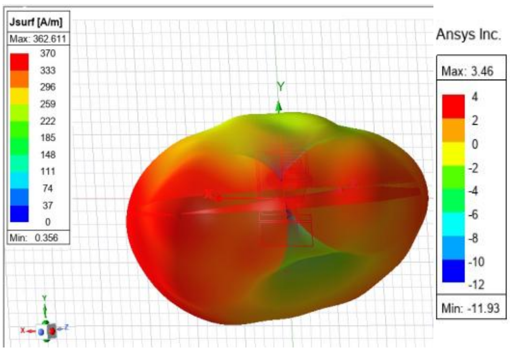
a) 1.79 GHz



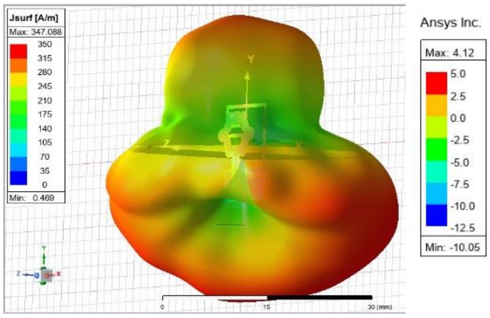
b) 3.84 GHz



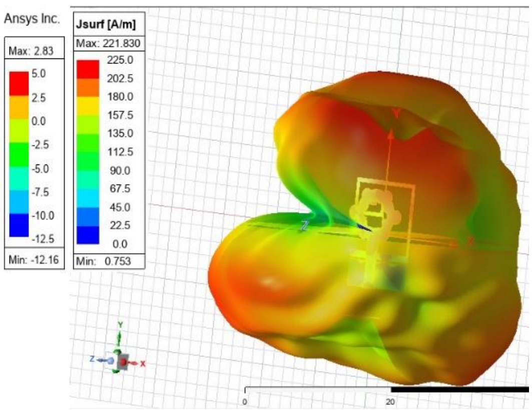
c)7.34 GHz



d)9.08 GHz



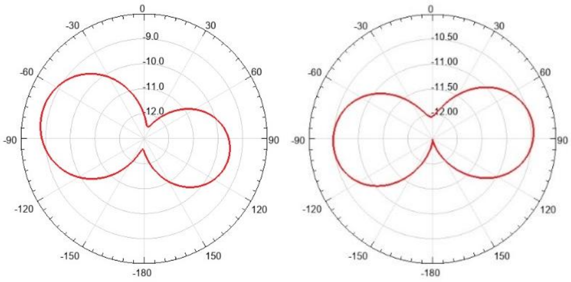
e) 11.44 GHz



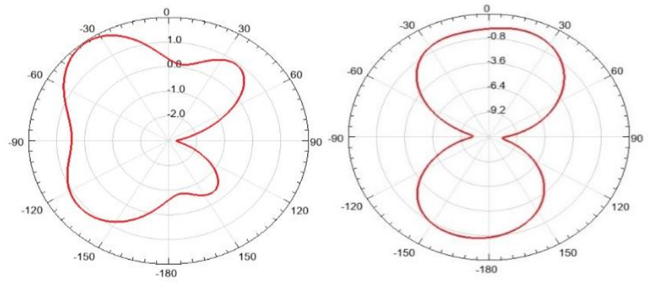
Gain of the antenna at frequencies: (a) 1.79GHz, (b) 3.84GHz, (c) 7.34GHz, (d) 9.08GHz and (e)11.44GHz.

4.6 Radiation Characteristics (E-Field and H-Field Patterns)

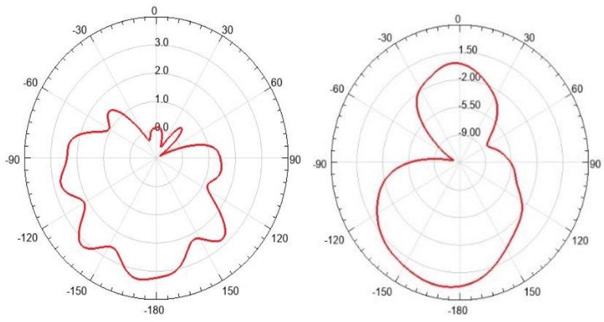
a) 1.79GHz



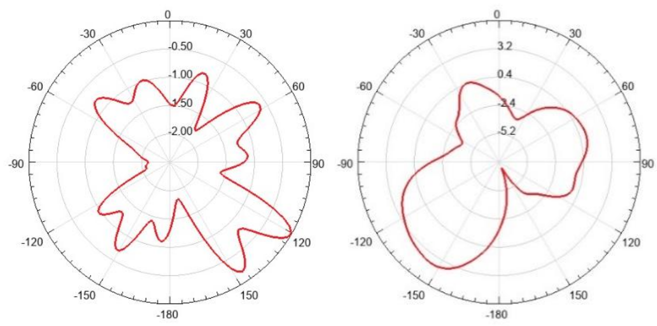
b)3.84GHz



c) 7.34GHz

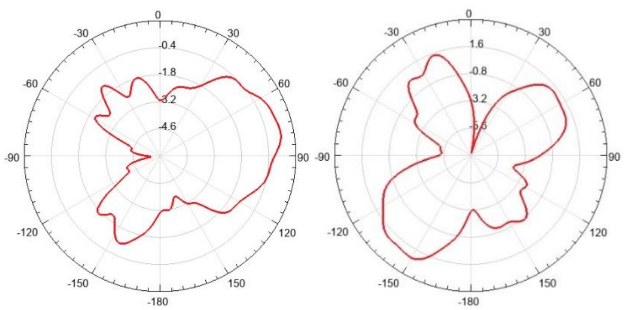


d) 9.08GHz



E-Field H-Field

e)11.44GHz



E-Field (left) and H-Field (right) patterns for the antenna at frequencies: (a) 1.79GHz, (b) 3.84GHz, (c) 7.34GHz, (d) 9.08GHz and (e)11.44GHz.

The simulated far-field radiation was examined in both the electric-field (E-plane) and magnetic-field (H-plane) planes. At the lower resonant bands (1.79 GHz and 3.84 GHz), the E-plane patterns were nearly omnidirectional, showing broad main lobes and smooth power distribution. The H-plane plots were symmetrical and stable, indicating uniform radiation and good impedance balance. At the mid and higher bands (7.34 GHz to 11.44 GHz), additional lobes appeared in both planes as higher-order modes became dominant due to the fractal geometry and the defected ground structure. The beamwidths narrowed slightly, but the main lobes remained oriented in the broadside direction. Overall, the antenna maintained consistent radiation characteristics across all resonant frequencies, confirming that the combined fractal and DGS design supports multiband operation without significant pattern distortion.

4.7 Discussion

The combination of fractal miniaturization and DGS clearly improved the antenna’s performance. Compared with a simple circular patch, the proposed design achieved multiple resonances with a much smaller footprint. The gain values and bandwidths are sufficient for most practical wireless systems, including GSM, 5G, radar, and satellite communication. Although the FR-4 substrate introduces some dielectric loss at high frequencies, the antenna still maintained good efficiency, typically above 80–85% across all bands. Overall, the results demonstrate that nature-inspired fractal patterns and DGS integration can be a practical approach to designing compact multiband antennas.

5. Conclusion

This project involved the modelling and simulation of a compact, flower-shaped fractal antenna from a published academic paper. A miniaturized multiband microstrip antenna employing a flower-shaped fractal radiating patch and a rectangular spiral defected ground structure (DGS) was designed, simulated, and tested. The design was replicated in ANSYS HFSS to analyze its multiband performance, which is achieved by combining fractal geometry with a defected ground structure.

The antenna was implemented on an FR-4 substrate with a dielectric constant of $\varepsilon_r = 4.4$, a loss tangent of 0.02, and a thickness of 1.6 mm. The final dimensions measured 15 × 11 × 1.6 mm^3, confirming substantial miniaturization compared to conventional microstrip designs. A 50 Ω microstrip feedline was used to excite the structure, and full-wave simulations were performed using ANSYS HFSS to evaluate the antenna’s electromagnetic characteristics.

The simulated results demonstrated six distinct resonant frequencies at approximately 1.79, 3.84, 7.34, 9.08, 11.44, and 14.6 GHz. All resonances exhibited reflection coefficients below −10 dB, indicating proper impedance matching and efficient radiation. The surface current analysis confirmed that the lower frequencies were governed by the fundamental TM_{11} mode, while higher bands arose from the interaction between the fractal petals and the spiral-shaped DGS. The radiation and gain plots verified that the antenna maintained stable performance across all bands, with a maximum realized gain of 6.58 dB and consistent broadside radiation patterns.

The study demonstrated that introducing fractal geometry on the radiating patch effectively extended the electrical length, enabling multiple resonant modes within a small physical area. The spiral DGS further enhanced performance by introducing additional reactive loading, improving bandwidth, and reducing return loss. The strong correlation between simulation and measurement validated the design methodology and confirmed the antenna’s suitability for GSM/UMTS, 4G/5G, C- and X-band radar, Ku-band, and satellite uplink applications.

6. References

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